Dynamic Size Limits Configuration

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Issued: July 2015

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1 Overview

This document provides the procedures to enable and use the Dynamic Size Limits feature in the ToolboxST* application. The procedures are as follows:

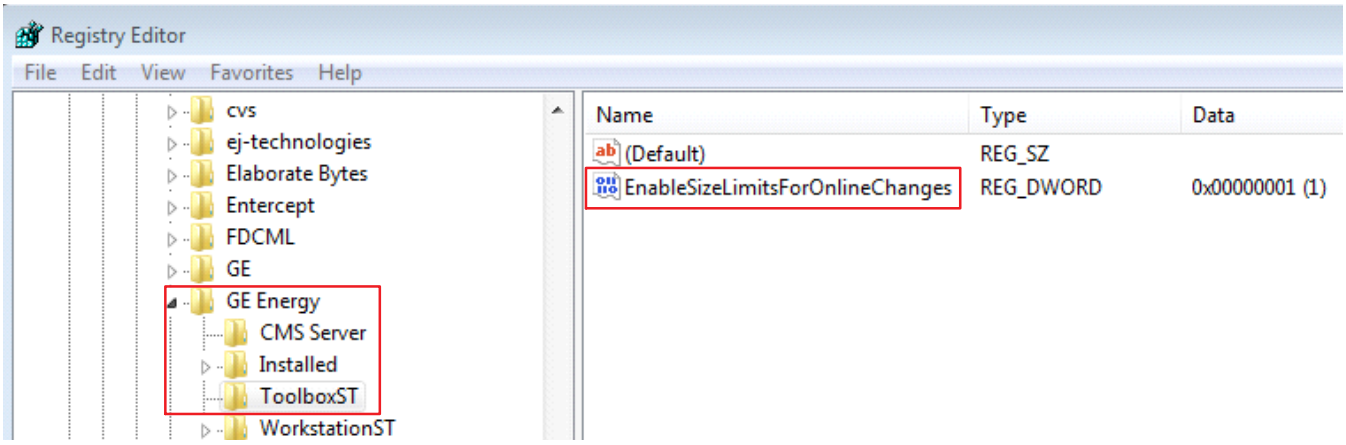
- *Enable Dynamic Size Limits*
- *Enable and Modify Send Value Options*
- *Color Code Bar Graphs*

2 Enable Dynamic Size Limits

A registry key must be added and configured to enable the Dynamic Size Limits feature.

➢ **To enable the Dynamic Size Limits feature**

1. Add the registry key *EnableSizeLimitsForOnlineChanges* with the value **1** at the following location:
   - For 64-bit machine: *HKEY_LOCAL_MACHINE\SOFTWARE\Wow6432Node\GE Energy\ToolboxST*
   - For 32–bit machine: *HKEY_LOCAL_MACHINE\SOFTWARE\GE Energy\ToolboxST*



Registry Key Added to Registry Location

2. This feature applies to variables with the following data types. Confirm that the variables are non-enumerated and fall within Work87gethese data types:

INT	UDINT
UINT	REAL
DINT	LREAL



Attention

Both steps in this procedure must be satisfied to enable the Dynamic Size Limits feature.

Note Restart the ToolboxST application if any changes are made to this registry.

3 Enable and Modify Send Value Options

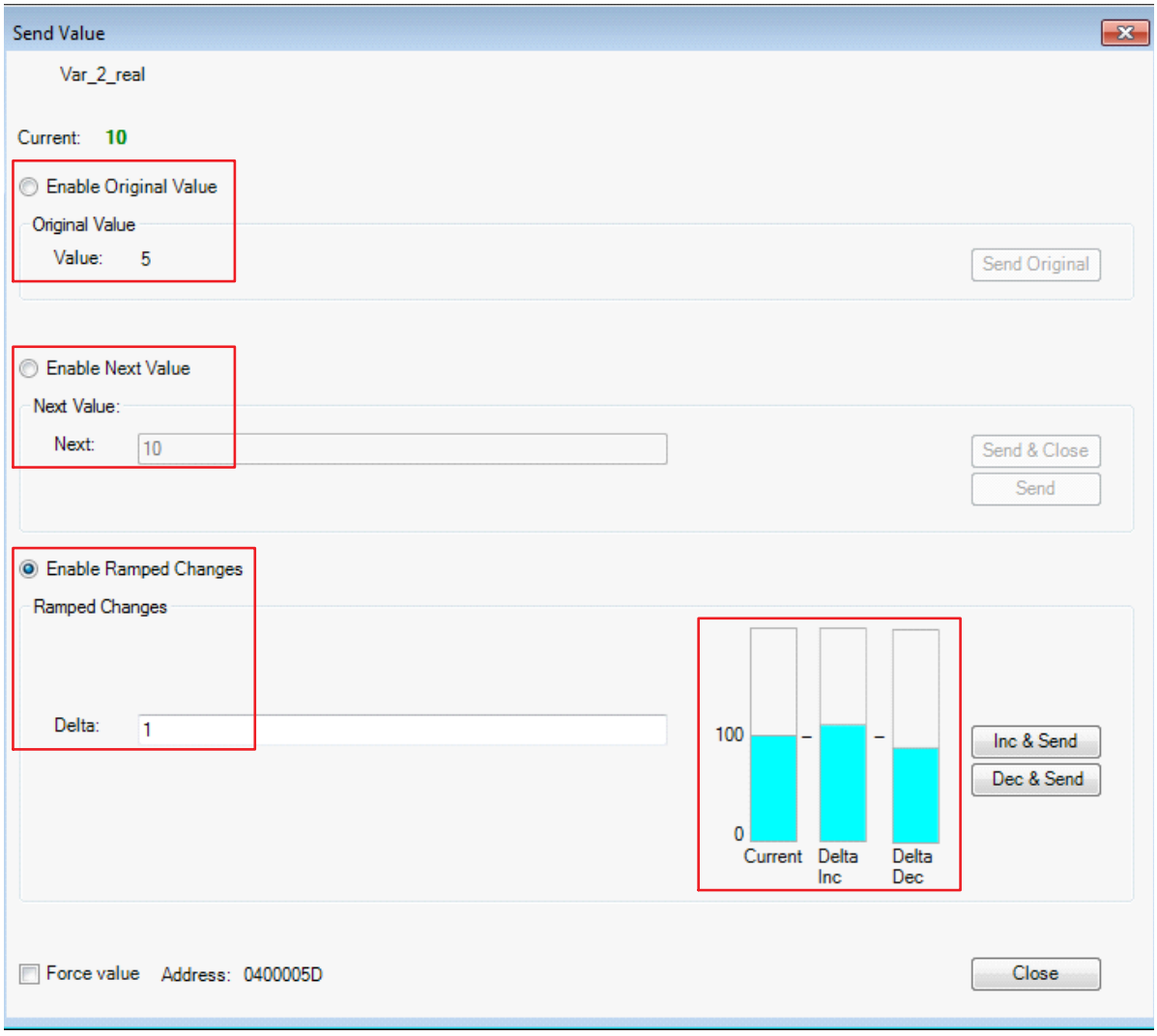
The existing *Send Value* dialog window has the following options to change live values:

- Original Value
- Next Value
- Ramped Change

➢ **To modify the *Send Value* option values**

From the *Send Value* dialog box, select an option and enter the appropriate live value.

Notes: The option **Enable Next Value** is the default selection. When one option is enabled, the other options are disabled.





Attention

If you are not authorized with live value modification privileges, these options are enabled but the buttons within each selection are disabled. For example, even if the Ramped Changes option is selected, the *Inc & Send* and *Dec & Send* buttons are disabled.

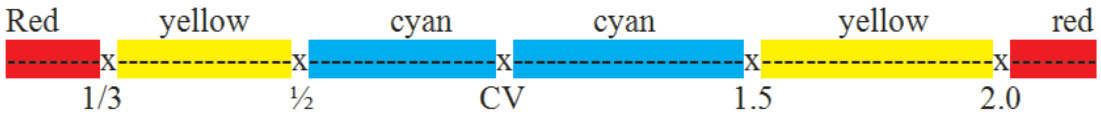
When the options **Original Value** or **Next Value** are selected, the *Send Value* dialog box displays two bar graphs to indicate their values, Current and Original and Current and Next, respectively. When the option **Ramped Changes** is selected, three bar graphs display to indicate the Current, Delta Inc, and Delta Dec values.

4 Bar Graph Color Codes

The size and color of the bar graphs on the *Send Value* dialog box are dynamically changed when the Current, Next, or Delta values are modified, and the bar graph values are stabilized between Current, Next, or Delta. For example, if the Current value is 100 and the Next value is 200, the Current bar graph value will be exactly half (1/2) of the value of the Next bar graph.

The first bar graph color is always *Cyan*. The subsequent bar graphs' colors are changed based on the following conditions:

CV: Current value



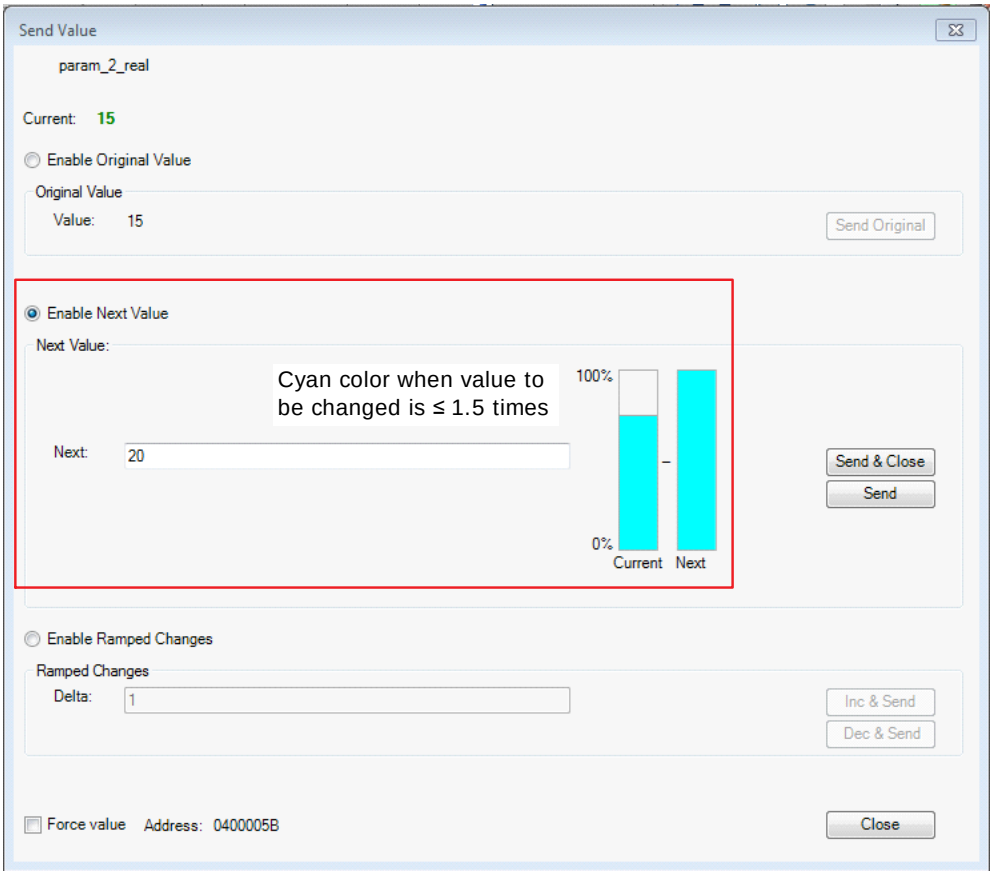
NV: Next value or Value to be changed

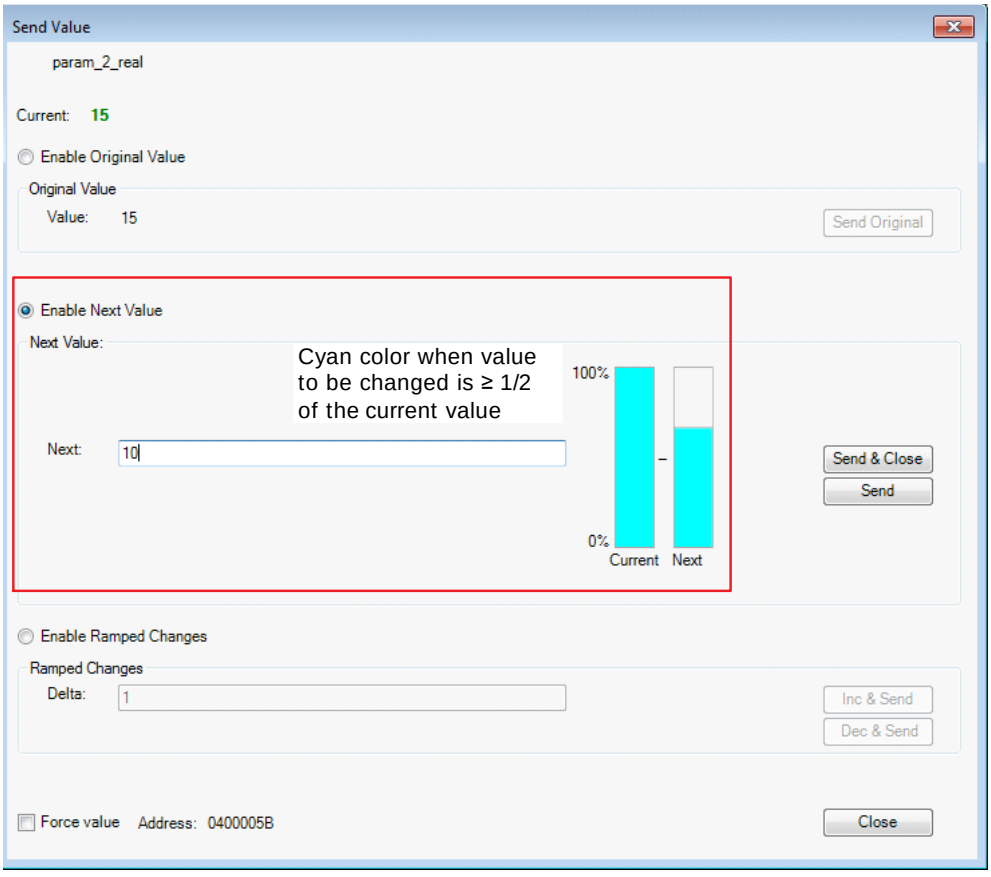
For example: Ramped Changes NV is CV+Delta for the Delta Inc bar graph and CV-Delta for the Delta Dec bar graph.

The following conditions determine the color of the Next or Delta Inc bar graph and the Delta Dec bar graph:

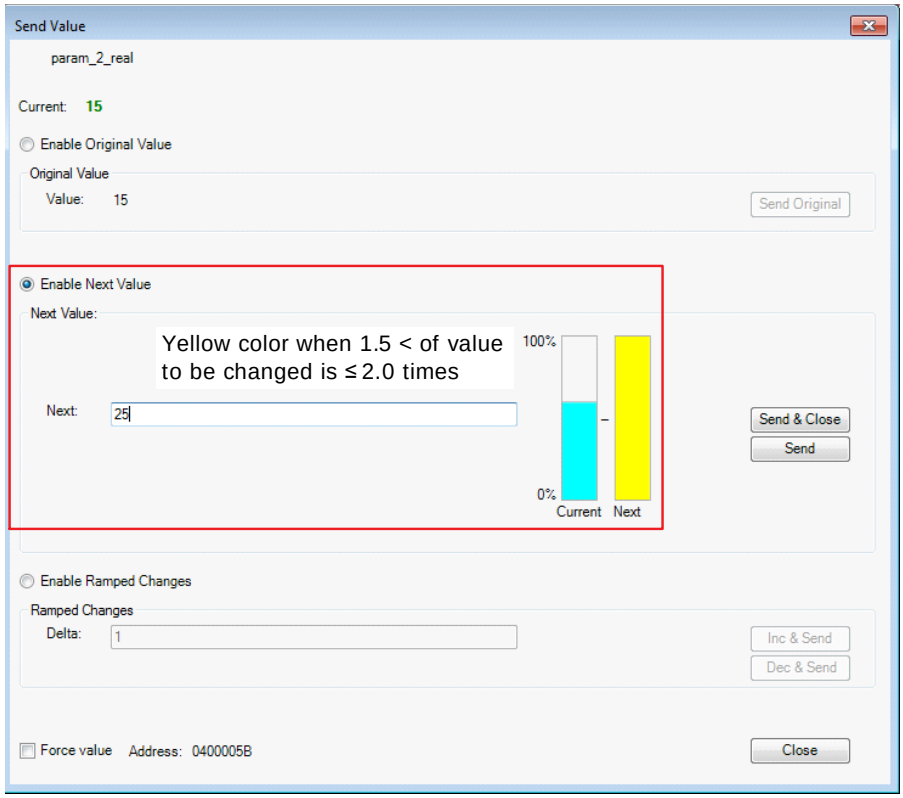
1. C-1: Display Current value or Value to be changed in the bar graph, or display the message Value to be changed with color (Cyan, Yellow, or Red).

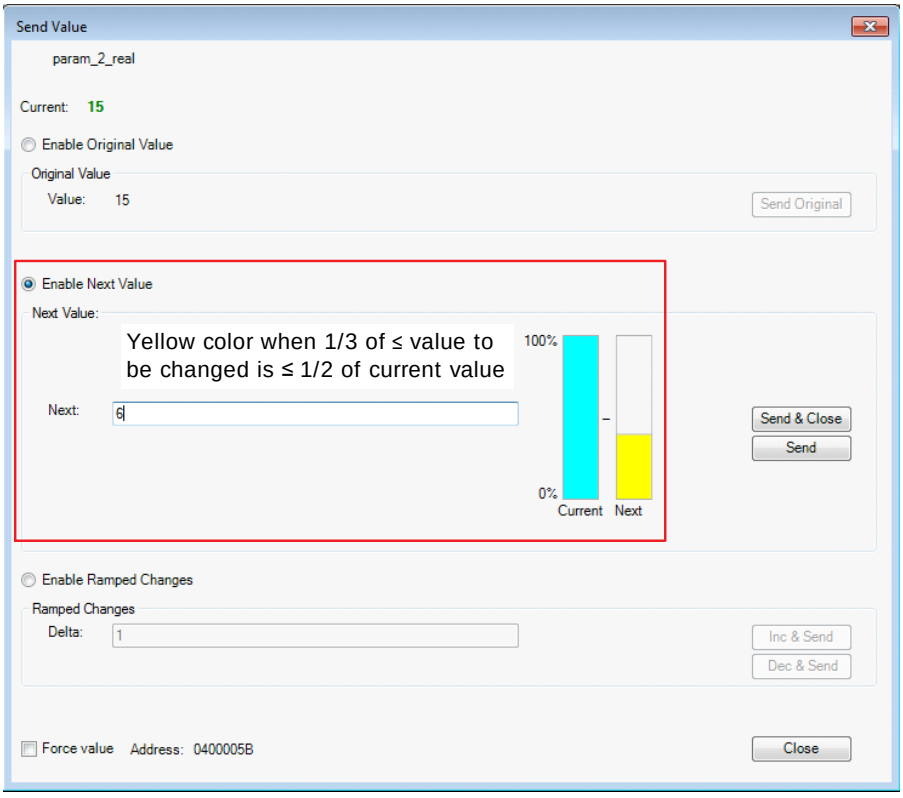
   a. C-1-I: Color is Cyan when Value to be changed is ≤ 1.5 times
   OR
   Value to be changed is ≥ 1/2 of the Current value.


Send Value
param_2_real
Current: 15
Enable Original Value
Original Value
Value: 15
Send Original
Enable Next Value
Next Value:
Cyan color when value to be changed is ≤ 1.5 times
Next: 20
100%
0%
Current
Next
Send & Close
Send
Enable Ramped Changes
Ramped Changes
Delta: 1
Inc & Send
Dec & Send
Force value
Address: 0400005B
Close

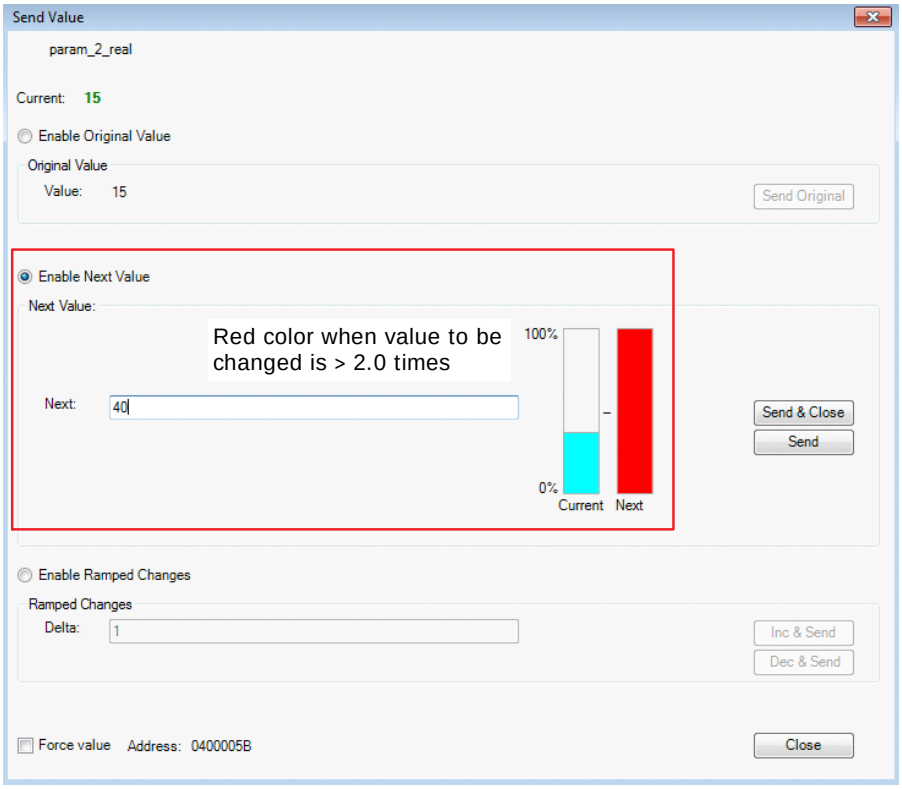

Send Value
param_2_real
Current: 15
Enable Original Value
Original Value
Value: 15
Send Original
Enable Next Value
Next Value:
Cyan color when value to be changed is ≥ 1/2 of the current value
Next: 10
100%
0%
Current
Next
Send & Close
Send
Enable Ramped Changes
Ramped Changes
Delta: 1
Inc & Send
Dec & Send
Force value
Address: 0400005B
Close

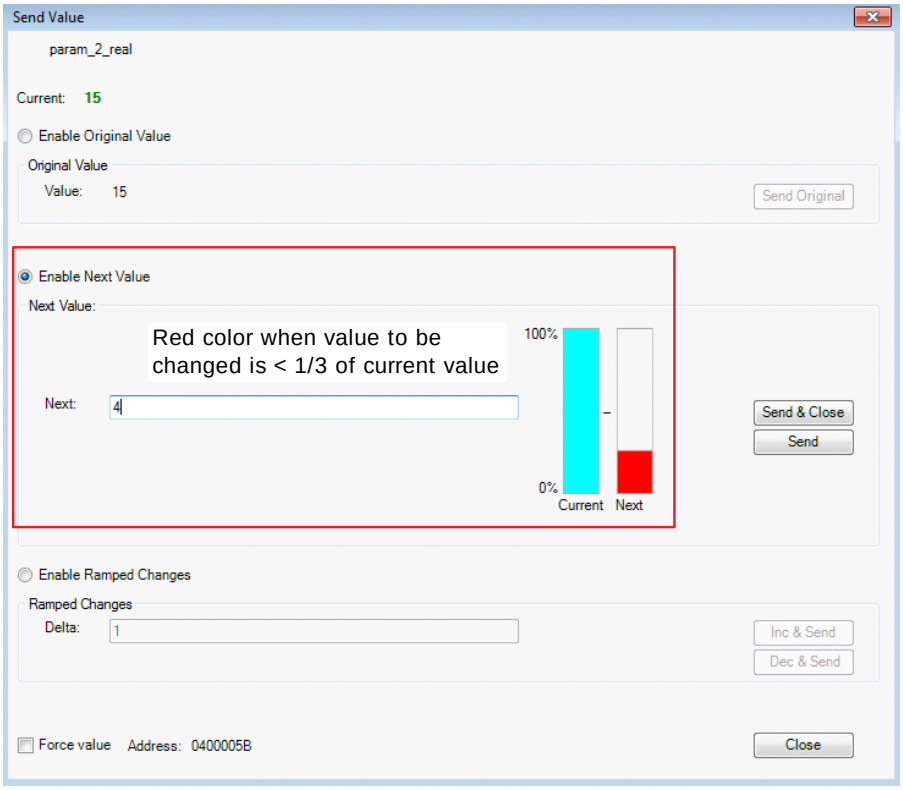
b. C-1-II: Color is Yellow when 1.5 is < Value to be changed and ≤ 2.0 times
OR
1/3 ≤ of the Value to be changed and < 1/2 of the Current value.



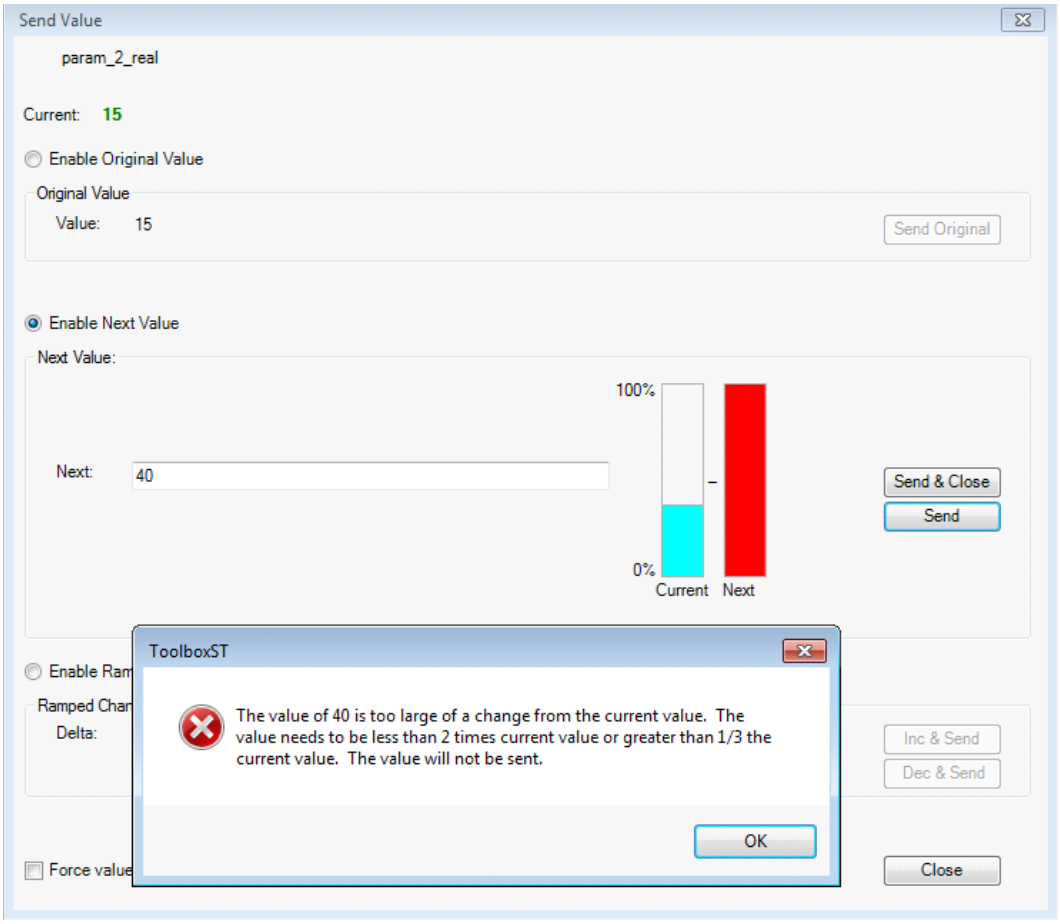


c. C-1-III: Color is Red when Value to be changed is > 2.0 times
OR
Value to be changed is < 1/3 of the Current value.



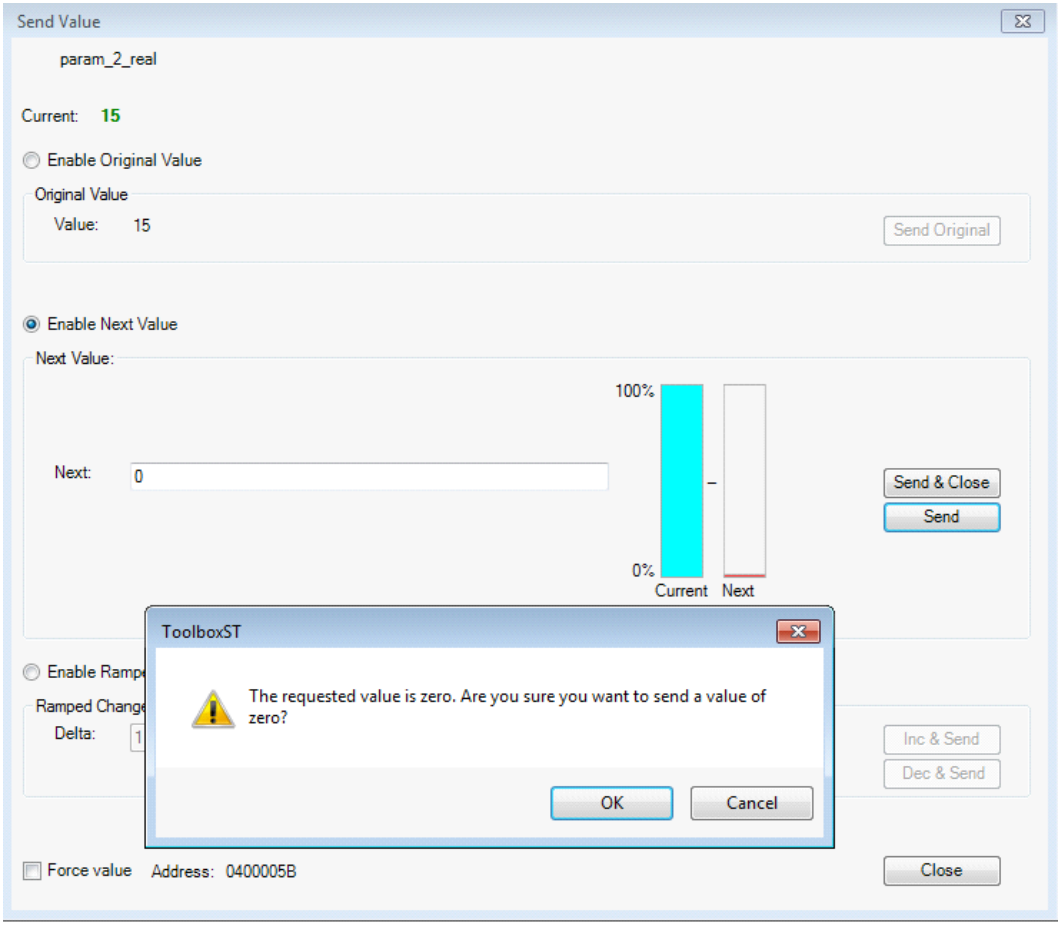


2. C-2: If Value to be changed falls within the C-1-III condition, the Value to be changed change will not be accepted.

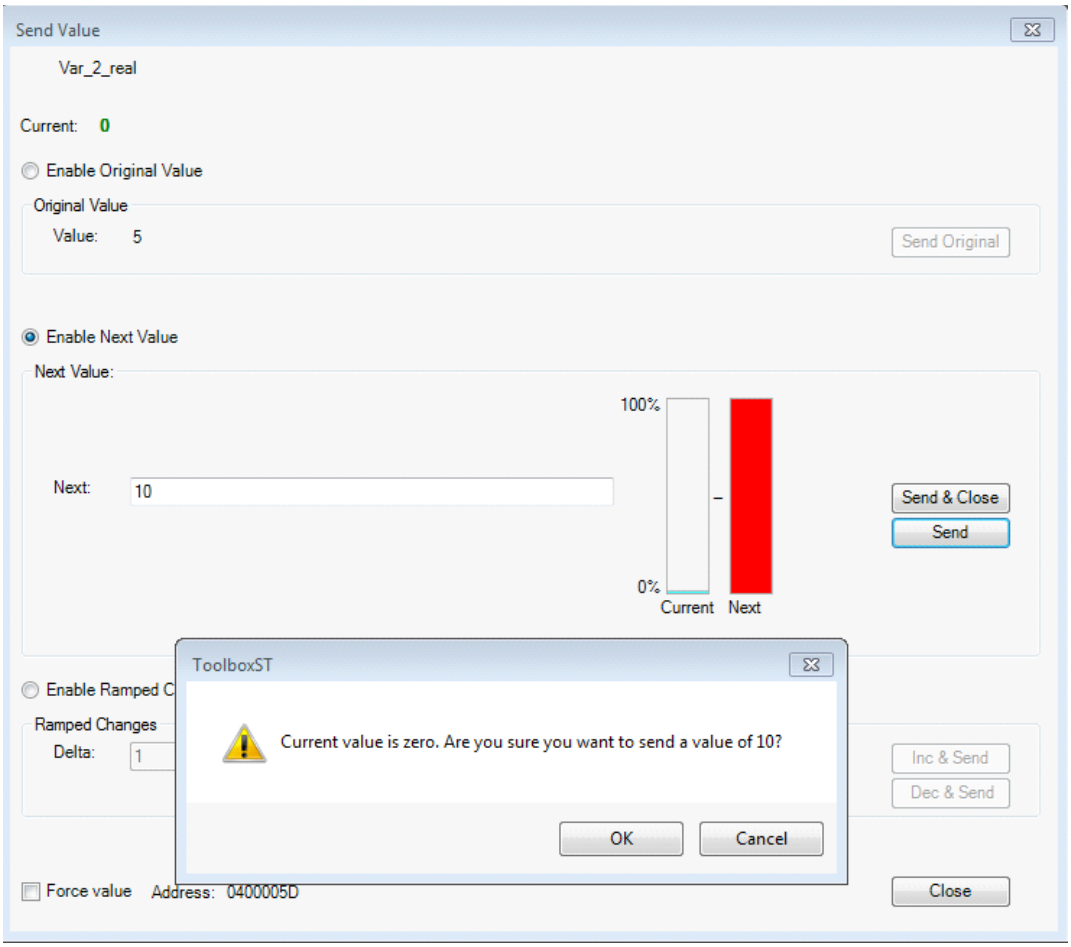


Value to Be Changed Too Large for Change

3. C-3: If the Value to be changed is set to 0 (zero), the *Send Value* dialog box displays a confirmation message before proceeding with the 0 value. If Current Value is set to 0 (zero), it displays a confirmation message before proceeding with any value entered for Value to be changed.



Value to Be Changed to Zero Confirmation Message



Current Value of Zero to Be Changed Confirmation Message

Note Dynamic size limit validation (not allowing the value and confirmation messages) occurs in addition to all other existing validations (such as Entry High and Entry Low limit validation).